AEL - product requirements document

Problem and users

- **Problem**: The platform streamlines the creation of a weekly handbook regarding AI adoption in sports retail supply chains.
- **Users**: The primary users are our research team members interacting with the tool via a terminal. The downstream user is the operations manager, Who reads the final handbook to make informed decisions about modernizing their inventory planning.

What it must let a user do

- The user must be able to input raw URLs or text files containing regional business data, SME digitalization reports, or interview transcripts.
- The platform will automatically extract relevant insights concerning demand forecasting and broken size runs, translate them into English, and format them into a structured Markdown file ready to be committed to our GitHub Pages wiki.

Output qualities

- The published handbook page must be written in clear, pragmatic English
- It must contain structured formatting (bullet points, clear paragraphs) focused on actionable financial and operational insights for the SME.
- Functional & Domain Accuracy
    - Sub-Sector Relevance: Every workflow, prompt template, and tool evaluation must directly address the operational realities of amateur teamwear: short-run sublimation printing, seasonal club orders (e.g., KNVB football/hockey season kick-offs), custom crest vectorization, and fabric waste reduction (cutting/nesting).
    - Maturity-Tiered Roadmaps: Content must prevent "pilot paralysis" by segmenting solutions into three explicit maturity tiers:
        - Tier 1 (Off-the-shelf): GenAI for marketing club packages, automated customer inquiry sorting.
        - Tier 2 (Workflow automation): AI-assisted pattern layout (fabric yield optimization) and vector cleaning for club crests.
        - Tier 3 (Predictive operations): Demand forecasting based on historical amateur league registration dates and inventory reorder alerts.

    - Quantifiable ROI Anchors: Every use case must include standard Dutch SME operational KPIs (e.g., cut-to-ship cycle time, fabric scrap reduction percentage, order turnaround during peak September/October season).
- Readability, Usability & Structure

Quality Attribute	Acceptance Criteria	Target Metric
Language & Accessibility	Dutch (B1/B2 business level) with English technical terms preserved (e.g., *prompting, pattern nesting, SKU*).	Flesch-Kincaid / Dutch readability equivalent accessible to non-technical operations managers.
Execution Speed	Every chapter must provide a 1-page "Quick-Start Action Sheet" for immediate deployment.	A team lead can execute a pilot test within 48 business hours of reading.
Scannability	0% dense theoretical jargon; strictly structured with decision trees, step checklists, and vendor evaluation matrices.	Max 2 text pages before a visual diagram, framework table, or code/prompt snippet.
Tool Agnosticism & Longevity	Focus on operational architecture rather than single volatile apps. Where tools are listed, evaluate by integration capability (ERP/CAD compatibility like Gerber/Lectra).	Workflows remain conceptually valid for at least 18–24 months without software dependency decay.

Known constraints

- The platform must be directed via an agentic Command Line Interface (CLI)
- It must output to our existing wiki/site on github
- Any team member, must be able to execute the pipeline.
- Every automated processing step must have a manual fallback.

(Deployment and reproducibility) Deployer:

- Running it is not going to require anyone to already know the command line. I will write a short setup doc with the exact commands to copy and paste, in order, so someone who has never opened a terminal before could still get from a raw source file to a published page without needing to understand what is happening underneath.
- I will pin our tool versions and dependencies so the pipeline behaves the same way on anyone's laptop, not just the machine it was built on.
- The manual fallback is not just a checkbox, if the CLI step for translation or formatting fails, or someone is not comfortable running it, there is a plain written procedure for doing that step by hand (for example, translating manually and dropping the text into the existing markdown template ourselves). So a bad day with the tool never blocks a page from getting published.
- Publishing itself goes through a normal commit-and-push to our GitHub Pages wiki, but nothing goes live without someone checking it first. This matches the course's own rule that "nothing reaches an SME reader unreviewed."

The source material is partly Dutch

- Local regional data and field interview recordings will be in Dutch. The platform itself must handle retrieval, translation, and critical appraisal autonomously.
- By building this translation step directly into the agentic workflow, any team member (including non-Dutch speaking exchange students) can initiate research queries. This ensures language barriers do not dictate who can contribute to the research.

Out of scope, for now

- Building a custom graphical user interface (GUI).
- Directly analyzing Deventrade's actual proprietary B2B sales data (the platform only processes research context for the handbook, not the company's live ERP data).

Open questions and assumptions

- Assumption*:* The agentic CLI can reliably maintain the context of the overarching research question across different prompt runs.
- Question: How effectively can the tool handle long, unstructured audio transcripts without hitting memory token limits or hallucinating facts? What AI tools is the company working with right now?